Derek Wang

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Education

University of Waterloo — **Expected Apr 2030**
Bachelor of Applied Science in Computer Engineering — GPA: 3.7

Experience

Autonomous Self Driving Software Engineer — **Sept 2025 – Present**
WATonomous — *Waterloo, ON, Canada*

- Coordinated collection and annotation of acomprehensive dataset of 3,448 images (3,087 train, 240 valid, 121 test) for object detection, implementing strategic data augmentation to ensure high-quality training data
- Trained YOLO11s object detection model using Ultralytics framework, achieving a 167% performance boost from 30% to 80% confidence rate through optimized hyperparameters and data augmentation strategies
- Exported trained model to ONNX format with onnxslim optimization for embedded deployment, creating a 36MB artifact ready for real-time inference on rover hardware

Autonomy Developer — **Aug 2025 – Present**
Waterloo Aerial Robotics Group — *Waterloo, ON, Canada*

- Developed Hardware-in-the-Loop (HITL) simulation systems in Python to address autonomous flight testing requirements for the 2026 National Annual UAS Student Competition
- Redesigned GPS movement architecture using periodic functions to eliminate unrealistic teleportation behavior, creating smooth trajectory interpolation systems that enhanced simulation realism and accuracy by 75%

Robotics Software Programmer – VEX Robotics Competition — **Sept 2024 – June 2025**
Checkmate Robotics Club — *Markham, Ontario*

- Programmed competition software for Team 16868C Rushdown in the VEX Robotics Competition, securing invitation to the 2025 VRC World Championships among top 10% of global teams
- Deployed advanced robotics systems using Okapi PROS Library, Odometry positioning, and PID Controllers, boosting robot performance by 25% through precise motion control and comprehensive error handling
- Designed and refactored autonomous navigation algorithms, creating 15-second match routes and 60-second programming skills routines that achieved 5th place out of 80 teams at Provincial Championships

Projects

Coffee Chat Scheduler | React | Python | FastAPI | PostgreSQL | Google Calendar API | Git | *Source Code* | *Live Demo*

- Built a full-stack web application using React and FastAPI, integrating Google Calendar API via OAuth 2.0 authentication with RESTful API endpoints, error handling, and CORS configuration
- Implemented real-time availability checking and automated calendar event creation, storing booking data in a PostgreSQL database with SQLModel Object Relational Mapping

Autonomous Vehicle Control System | C++ | Docker | CMake | Foxglove | *Source Code* | *Demo Video*

- Engineered LiDAR-based perception pipeline processing sensor data to generate real-time costmaps with 0.1m resolution, integrating persistent memory mapping system for 95% accurate dynamic environment representation
- Orchestrated autonomous navigation stack featuring A* path planning algorithm and Pure Pursuit tracking algorithm, achieving sub-0.5m waypoint accuracy and 100% obstacle avoidance success rate across 50+ test scenarios

Fruit Ninja AI | Python | YOLO11 | *Source Code*

- Architected and implemented an automated computer vision pipeline using YOLOv11 object detection, achieving 90% accuracy in fruit classification
- Built comprehensive dataset of 816 images (514 training, 302 testing), optimizing model performance through strategic data augmentation and validation techniques

Skills

Languages: C++, C, Java, Python, HTML5, CSS, JavaScript, C#, TypeScript, SQL
Engineering Tools: Git, Fusion 360, Docker, Foxglove, Unity, PostgreSQL, MongoDB, Supabase
Technologies: React, FastAPI, Node.js, Express, PyTorch, OpenCV, Ultralytics, Tailwind CSS
Certifications: Docker Foundations Professional